

BAREILLY, IndiaWMO: **421890**Lat: **28.422N**Lon: **79.451E**Elev: **177**StdP: **99.22**Time Zone: **5.50 (IND)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.9	9.0	4.3	5.3	15.7	5.8	5.8	15.3	2.5	16.0	2.4	15.8	0.5	270	0.318

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	11.3	41.1	23.9	39.7	24.3	38.4	24.4	29.2	34.0	28.7	33.3	28.4	32.8	1.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	24.8	31.3	27.7	24.2	31.0	27.3	23.7	30.7	97.5	34.5	95.4	33.5	93.6	33.0	31.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3.3	2.5	2.2	4.9	43.5	2.6	1.1	3.0	44.2	1.5	44.9	0.0	45.5	-1.9	46.3
			4.3	30.2	2.7	0.7	2.4	30.7	0.9	31.1	-0.6	31.5	-2.6	32.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.6	14.8	19.2	24.4	30.1	32.2	32.2	29.7	29.8	29.2	26.6	21.8	16.4	
	DBStd	6.36	2.65	2.58	3.08	2.76	2.38	2.77	1.95	1.63	1.87	2.06	1.95	2.81	
	HDD10.0	2	1	0	0	0	0	0	0	0	0	0	0	1	
	HDD18.3	203	113	18	1	0	0	0	0	0	0	0	1	70	
	CDD10.0	5678	150	257	446	602	689	667	611	614	575	514	353	200	
	CDD18.3	2837	3	42	188	352	431	417	352	356	325	256	103	11	
	CDH23.3	35626	10	227	1712	4748	6485	6271	4606	4660	3977	2381	516	34	
CDH26.7	18938	0	38	720	2831	4131	3939	2217	2244	1799	934	84	1		
Wind	WSAvg	0.6	0.4	0.7	0.7	0.8	0.9	0.9	0.6	0.5	0.5	0.3	0.2	0.3	
Precipitation	PrecAvg	1155	23	27	12	11	21	120	353	307	196	40	3	7	
	PrecMax	1551	78	99	91	64	81	374	582	590	529	219	27	35	
	PrecMin	633	3	0	0	0	0	0	34	77	6	14	0	0	
	PrecStd	289	21	27	20	14	25	111	146	137	138	41	7	10	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.6	30.2	36.3	41.0	42.2	43.0	37.5	36.1	35.6	34.1	29.9	25.9	
		MCWB	17.8	20.4	23.3	23.3	23.7	24.4	28.0	28.1	27.1	24.9	20.5	18.0	
	2%	DB	22.9	27.8	34.1	39.6	41.0	41.4	35.9	34.9	34.4	32.9	28.7	24.2	
		MCWB	16.9	19.4	22.1	23.1	23.7	24.3	28.2	28.1	27.0	24.4	20.3	17.4	
	5%	DB	21.4	26.1	32.3	38.3	39.7	39.9	34.6	34.0	33.6	32.0	27.7	22.9	
		MCWB	15.9	18.6	21.3	23.0	23.8	25.4	28.2	28.0	26.9	24.3	20.0	17.0	
	10%	DB	19.9	24.6	30.7	36.9	38.3	38.1	33.5	33.2	32.8	31.0	26.5	21.5	
		MCWB	15.0	17.9	20.7	22.6	24.0	26.0	28.1	27.8	26.8	23.9	19.5	16.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.0	21.7	25.4	28.6	29.5	29.5	29.6	29.3	29.0	27.2	22.6	19.4
			MCDB	23.1	28.0	33.7	37.2	35.9	35.3	34.0	32.8	32.8	31.2	27.1	24.1
2%		WB	17.7	20.3	23.8	27.0	28.3	28.8	29.1	28.8	28.2	26.7	21.7	18.5	
		MCDB	21.9	26.0	32.2	35.6	34.8	35.1	33.4	32.8	31.9	30.7	26.4	22.7	
5%		WB	16.6	19.5	22.5	25.4	27.2	28.3	28.7	28.5	27.8	26.0	21.0	17.6	
		MCDB	20.3	24.8	29.7	34.3	34.9	34.3	32.8	32.3	31.5	30.1	25.9	21.6	
10%		WB	15.7	18.6	21.5	24.1	26.2	27.8	28.3	28.2	27.4	25.1	20.3	16.8	
		MCDB	18.9	23.5	28.5	34.0	34.2	33.6	32.1	31.8	31.1	29.3	25.0	20.5	
Mean Daily Temperature Range	5% DB	MDBR	9.1	10.5	11.6	13.0	11.3	9.1	5.7	5.6	6.4	9.3	10.4	9.5	
		MCDBR	11.3	11.8	12.9	13.7	13.0	12.0	7.6	7.0	7.6	9.4	10.8	11.2	
		MCWBR	6.1	5.2	4.6	4.9	4.2	3.8	2.6	2.4	2.7	3.3	4.3	5.5	
	5% WB	MCDBR	10.2	10.9	12.1	12.5	10.3	8.6	6.6	6.1	6.6	7.6	9.9	9.9	
		MCWBR	6.1	5.4	5.3	6.2	4.6	3.5	2.8	2.6	2.9	3.3	4.3	5.2	
Clear-Sky Solar Irradiance	taub	0.513	0.508	0.504	0.575	0.678	0.735	0.709	0.627	0.583	0.635	0.612	0.557		
	taud	1.811	1.880	1.924	1.795	1.630	1.567	1.638	1.826	1.887	1.685	1.670	1.728		
	Ebn at Noon	696	752	789	747	674	633	648	700	715	632	600	627		
	Edh at Noon	183	185	188	220	260	276	256	211	193	222	210	192		
All-Sky Solar Radiation	RadAvg	2.73	4.19	5.61	6.51	6.71	5.73	4.53	4.65	4.74	4.50	3.56	2.87		
	RadStd	0.48	0.34	0.23	0.24	0.34	0.65	0.41	0.50	0.48	0.24	0.21	0.24		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A